

F-TALENT CODE 2026

Dynamic Path Maximum Queries on Tree

Time limit: 1.00 s | Memory limit: 512 MB

Problem Statement

In many real-world systems such as computer networks, hierarchical organizations, or transportation graphs, data is naturally organized as a tree. Efficiently tracking and updating critical values along paths between nodes is essential for performance monitoring, resource allocation, and decision-making. The problem of Dynamic Path Maximum Queries on Tree addresses exactly this need: it requires maintaining the maximum value on any path while supporting frequent updates to individual nodes.

You are given a tree consisting of n nodes, numbered from 1 to n . Each node holds a numerical value. Your task is to process a sequence of queries of two types: update the value of a specified node s to a new value x , or find the maximum value appearing on the unique path connecting two nodes a and b .

Input

- The first input line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$
- The next line has n integers $v_1, v_2, \dots, v_n$: the value of each node.
- Then there are $n-1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .
- Finally, there are q lines describing the queries. Each query is either of the form " $1\ s\ x$ " or " $2\ a\ b$ ".

Output

Print the answer to each query of type 2.

Constraints

$$1 \leq n, q \leq 2 \cdot 10^5$$

$$1 \leq a, b, s \leq n$$

$$1 \leq v_i, x \leq 10^9$$

Example 1:

INPUT	OUTPUT
5 3 2 4 1 3 3 1 2 1 3 2 4 2 5 2 3 5 1 2 2 2 3 5	4 3

Example 2:

INPUT	OUTPUT
10 10 9 2 1 1 1 4 2 10 7 4 2 1 3 2 4 2 5 4 6 5 7 6 8 4 9 8 10 2 2 5 4 1 10 4 1 5 9 2 5 4 2 9 5 2 1 10 2 1 6 1 8 4 1 3 5 2 6 1	1 9 10 9 9 9